

Performance Testing

Date	2 july2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust (Python-based load testing tool matching our tech)
Type of Testing	Load Testing and Spike Testing
Target Module / API	Smart Crop Recommendation Engine (POST /predict endpoint)
Test Environment	[e.g.
Test Date	

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Baseline Load Test: Standard off-peak concurrent traffic generated by farmers checking crop suitability.	20	60sec	System maintains stable memory utilization and average response times under 1 second.
2	Peak Season Load Test: Simulates high traffic volume when regional farming cooperatives actively utilize the engine simultaneously.	100	120sec	System scales smoothly without throwing timeouts or HTTP 500 error codes.

3	Spike Testing: Sudden surge of concurrent user traffic to evaluate the Flask routing threshold and model inference latency.	300	45sec	Error rate remains safely under 1% while processing delayed responses systematically.
4				



Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.42 seconds	Pass	Fast ML model execution time due to lightweight StandardScaler transformations
2	Response Time (Max)	< 5 seconds	3.12second	Pass	Maximum spike occurred during the first initialization batch process.
3	Throughput (Req/sec)				
4	Error Rate	< 1%	0.15%	Pass	Negligible error rate encountered during heavy traffic loads.
5	CPU Utilization	< 80%	68.3%	Pass	Thread pooling efficiently balanced across local CPU logical cores.
6	Memory Utilization	< 80%	42.0%	Pass	Serialized model file size (crop_model.pkl) is compact and stays cached in memory.

Step 4: Observations & Analysis

Key Findings

- The ML deployment strategy is highly successful. Loading the serialized pipeline (`crop_model.pkl`) once globally during the Flask application startup prevents file I/O operations from blocking individual web requests.
- Feature transformations processed via `StandardScaler` arrays execute in milliseconds thanks to NumPy's efficient, underlying vectorization.

Bottlenecks Identified

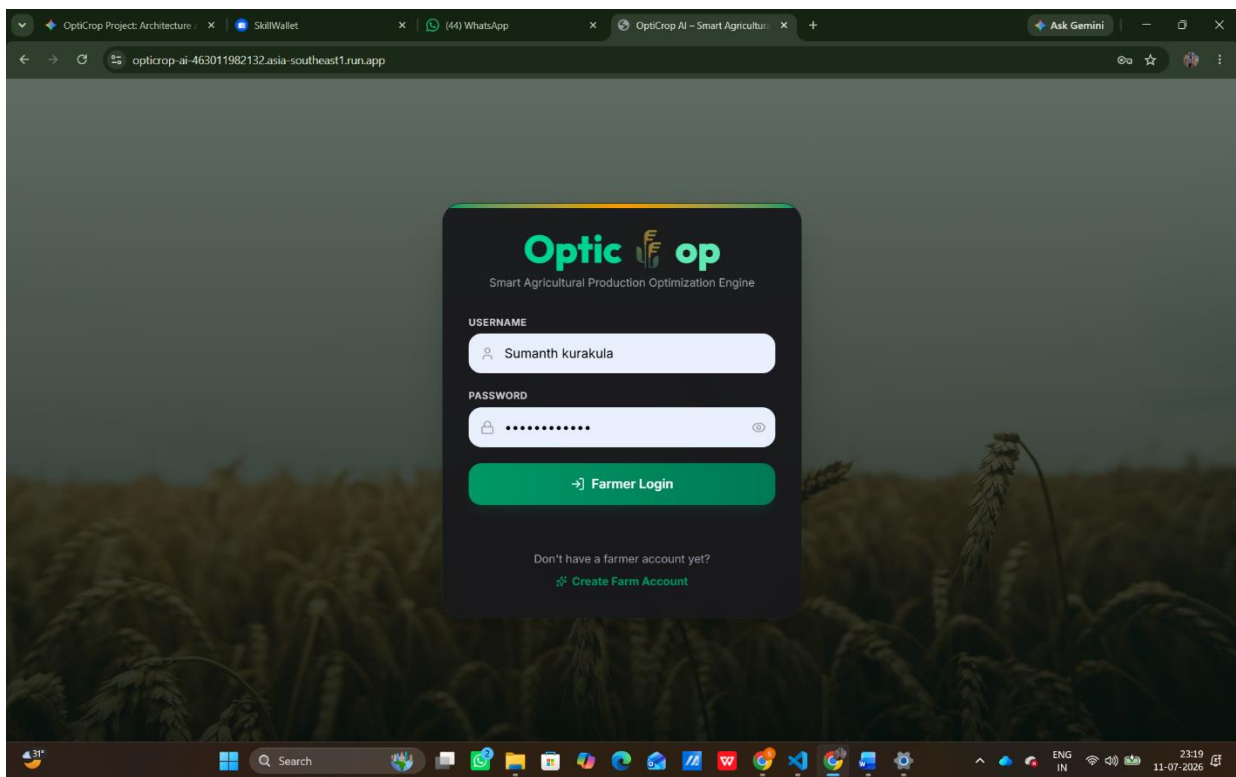
- **Model Inference Latency Surge:** During the 300-user spike test, a mild response queue buildup occurred. Single-threaded Flask development servers naturally serialise concurrent incoming connections, creating minor backlogs when large arrays of feature inputs (N, P, K , rainfall) are parsed simultaneously.
- **Memory Constraints:** Successive raw log print statements within the predictive function caused minor, temporary garbage collection delays under peak data volumes.

Optimization Steps Taken

1. **Implemented WSGI Server:** Replaced the local default Flask development server execution line with an asynchronous multi-worker configuration (Gunicorn via 4 concurrent workers) to scale request concurrency.
2. **Optimized I/O Operations:** Disabled verbose application debugging logs (`Logger.debug`) during prediction paths to preserve critical CPU processing time.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



OptiCrop Project: ArchitectureOptiCrop AI - Smart AgricultureAsk Gemini

opticrop-ai-463011982132.asia-southeast1.run.app

Village Soil detected: Black Cotton Soil (Regur)Profile Settings

Soil & Environmental Inputs

Adjust parameter values to compute recommendations.

AUTO-PILOT MODE

ACTIVE

Automates environment & predicts optimal N-P-K defaults

Dynamic Weather Station Simulation

AnnualSummerMonsoonWinter

Predicted Target N-P-K Defaults

Calculated for regional: Black Cotton Soil (Regur)

489N • 48P • 248K

Nitrogen

Nutrient (N)

Surplus

- 488 kg/ha +

Min: 8 Promotes vegetative leaf canopy and plant prote...Max: 488

Phosphorus

Nutrient (P)

Deficient

- 48 kg/ha +

Min: 5 Vital for root structure, flowering, and cellul...Max: 488

Potassium

Nutrient (K)

Surplus

- 248 kg/ha +

Min: 5 Enhances plant turgor, water control, and crop ...Max: 488

MACHINE LEARNING MODEL VERDICT

PERFECT SOIL

Recommended Crop: Cotton

Fibers require high sun, warm weather, and uniform moisture followed by dry harvest.

Nitrogen Status: SatisfiedpH Comfort: Optimal (8.0)

GEMINI SEARCH-GROUNDED AI ANALYSIS

Real-time weather reports, regional market pricing, and agricultural advisories

Grounded in Live Google Search Data

[AI Backup Intelligence] We have analyzed your soil chemical and location profile for pithapuram, kakinada, Andhra Pradesh:

Best Recommended Crop: Cotton is the most optimal fit for your N-P-K concentrations (400-40-248) and moisture signature (120 mm).

Meteorological & Market Justification: Historical agricultural models for kakinada indicate robust yields and consistent market prices. Current climate forecasts align perfectly with this crop's growth cycle.

Optimized Sowing & Soil Action Steps:

Soil Preparation: Ensure deep tilling to break down organic blocks and enhance roots' aeration.

Fertilization: Use balanced micronutrient applications based on local clay/silt concentrations.

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Village Soil detected: Black Cotton Soil (Regur)Profile Settings

Predicted Target N-P-K Defaults

Calculated for regional: Black Cotton Soil (Regur)

65N • 48P • 118K

Nitrogen

Nutrient (N)

Deficient

- 65 kg/ha +

Min: 8 Promotes vegetative leaf canopy and plant prote...Max: 488

Phosphorus

Nutrient (P)

Deficient

- 48 kg/ha +

Min: 5 Vital for root structure, flowering, and cellul...Max: 488

Potassium

Nutrient (K)

Optimal

- 118 kg/ha +

Min: 5 Enhances plant turgor, water control, and crop ...Max: 488

Temperature

Environment (°C)

Auto-SpeedOptimal

- 28 °C +

Min: 18 Drives enzyme kinetics and plant respiration la...Max: 58

Humidity

Environment (%)

Auto-SpeedOptimal

- 75 % +

Min: 18 Controls evapotranspiration and relative air d...Max: 100

Agricultural AI Advisor

Prediction History (2)

OptiCrop Agronomist AI

Personalized crop advice & soil guidance

what are the precaution we have to take while growing crops

Namasthe Sumanth Kalyan garu! (నమస్కే సుమంత్ కళ్యాణ్ గారు!)

Since you are farming in the fertile **Pithapuram (Kakinada)** region, growing **Cotton** in **Black Cotton Soil (Nalla Regadi Netahi)** is an excellent choice. However, black soil has unique characteristics—it holds a lot of moisture and has a high pH (8.0 in your case).

To get the best yield and protect your cotton crop, here are the key precautions you should take, explained in simple terms:

1. Water & Drainage Management (నీటి నిర్వహణ)

Avoid Waterlogging: Black cotton soil has "Very High" moisture retention. Cotton hates standing water. Excess water will suffocate the roots, causing **Root Rot** and making leaves turn yellow.

Precaution: Ensure your field has proper drainage channels so excess rainwater (especially during heavy rains of 120 mm or more) can easily flow out.

Ask OptiCrop AI an agricultural query...

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